Roll No.: BL.AI.U4AID24140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – November 2025

Third Semester

Artificial Intelligence & Data Science

23AID206 Introduction to Computer Networks

Duration: 2 hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Analyze the requirements for a given organizational structure to select the most appropriate networking architecture and technologies.
CO02	Analyze the working of protocols in the internet protocol stack for network applicatior.s.
CO03	Configure a router using simulator/emulator.
CO04	Analyze the network data to detect potential security threats in a network .

Answer all questions.

Be precise and to the point while answering questions.

1 a. Explain the key components of a computer network. How does each component contribute to network functionality?

[5] [CO01] [BTL 2]

b. Given a small organization with 20 computers in a single building, suggest a suitable network topology and justify your choice

[5] [CO01] [BTL 3]

2 Analyze which networking architecture (LAN, MAN, WAN) would best suit a bank with multiple city branches, and explain why?

[10][CO01] [BTL 4]

3 a. What is an IP Address? Perform a comparative analysis of IPv4 & IPv6

[5][CO02][BTL 1]

b. Explain how IP addresses are classified into different classes based on their range and intended use. [5] [CO02][BTL 2]

4 Explain the TCP/IP Model in detail. [10] [CO02] [BTL 2]

5 a. What is a URL? What do each of the 4 parts of a URL signify? Explain.

[5] [CO02] [BTL 2]

b. Explain the structural components of a World Wide Web. [5] [CO02] [BTL 2]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**20**	BTL 1	**5**
CO02	**30**	BTL 2	**30**
CO03	-	BTL 3	**5**
CO04	-	BTL 4	**10**